🔬 Continue Lab 2

Step 5: After changing sga_max_size to 8G

1. Error when startup again

```
SQL>  alter system set sga_max_size=8G scope=spfile;

System altered.

SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup
ORA-27104: system-defined limits for shared memory was misconfigured
SQL> 
```

2. Solution

creation of pfile , edit the value in it

```
SQL> create pfile from spfile;

File created.

SQL> !ls $ORACLE_HOME/dbs
hc_SADB.dat  init.ora  initSADB.ora  lkSADB  orapwSADB  spfileSADB.ora

SQL> !vim  $ORACLE_HOME/dbs/initSADB.ora
```

```
SQL> !cat $ORACLE_HOME/dbs/initSADB.ora | grep sga_max_size
*.sga_max_size=2g
```

3. Startup by pfile and overwrite spfile

```
Connected to an idle instance.

SQL> startup pfile='$ORACLE_HOME/dbs/initSADB.ora'
ORACLE instance started.

Total System Global Area 2147481656 bytes
Fixed Size                  8898616 bytes
Variable Size             486539264 bytes
Database Buffers         1644167168 bytes
Redo Buffers                7876608 bytes
Database mounted.
Database opened.
SQL> create spfile from pfile;

File created.

SQL> show parameter pfile

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
spfile                               string
SQL>
```

4. Startup again

```
SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup
ORACLE instance started.

Total System Global Area 2147481656 bytes
Fixed Size                  8898616 bytes
Variable Size             486539264 bytes
Database Buffers         1644167168 bytes
Redo Buffers                7876608 bytes
Database mounted.
Database opened.
SQL> show parameter pfile

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
spfile                               string      /u01/app/oracle/product/19.0/d
                                                 bhome_1/dbs/spfileSADB.ora
SQL>
```

🔬 Lab 3

Step 1: shutdown "SADB" with the most safe and fast method.

- Safe but will take a long time --> normal , transactional
- Fast but not safe --> abort
- Then, most suitable way for safe and fast shutdown --> immediate

```
SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
```

Step 2: Startup "SADB" step by step.

```
SQL> startup nomount;
ORACLE instance started.

Total System Global Area 2147481656 bytes
Fixed Size                  8898616 bytes
Variable Size             486539264 bytes
Database Buffers         1644167168 bytes
Redo Buffers                7876608 bytes
SQL> alter database mount
  2  ;

Database altered.

SQL> alter database open ;

Database altered.
```

Step 3: Show

1. Show the version of your database using the view "v$version"

```
SQL> select banner from v$version;

BANNER
--------------------------------------------------------------------------------
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
```

2. Show hostname, instance name, status, open mode and startup_time using the views "v$instance" and "v$database"

```
SQL> SELECT i.host_name, i.instance_name, i.status, d.open_mode, i.startup_time
 from v$instance i, v$database d;

HOST_NAME
----------------------------------------------------------------
INSTANCE_NAME    STATUS       OPEN_MODE            STARTUP_T
---------------- ------------ -------------------- ---------
oraclevm
SADB             OPEN         READ WRITE           11-JUN-26
```

3. Show the name and type of parameter "db_create_file_dest" using view "v$parameter"

```
SQL> select name, type from  v$parameter where name = 'db_create_file_dest';

NAME
--------------------------------------------------------------------------------
      TYPE
----------
db_create_file_dest
         2
```

Step 4: Create PERMANENT tablespace

- tablespace with name "iti_data" with two data files with intial size 50 MB each, 250 MB maxsize and increment of 10 MB

```
SQL> create tablespace iti_data datafile '/u01/app/oracle/oradata/SADB/iti_data_01.dbf' size 50M autoextend o
n next 10M maxsize 250M, '/u01/app/oracle/oradata/SADB/iti_data_02.dbf' size 50M autoextend on next 10M maxsi
ze 250M;

Tablespace created.
```

Step 5: Show using the views "dba_users" and "DBA_ROLE_PRIVS"

- show the usernames, their passwords, their tablespace, their roles and their acccount status using the views "dba_users" and "DBA_ROLE_PRIVS"
- left join was used to return all users from dba_users, even if they don’t have any roles in dba_role_privs.

```
SQL> select u.username, u.password, u.default_tablespace, r.granted_role , u.account_status from dba_users u
left join dba_role_privs r on u.username = r.grantee ;

USERNAME
--------------------------------------------------------------------------------
PASSWORD
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE
------------------------------
GRANTED_ROLE
--------------------------------------------------------------------------------
ACCOUNT_STATUS
--------------------------------
SYS

SYSTEM

USERNAME
--------------------------------------------------------------------------------
PASSWORD
--------------------------------------------------------------------------------
DEFAULT_TABLESPACE
```

Step 6: Show using view "dba_data_files".

- show data files names, thier tablespaces, their size, their status using view "dba_data_files"

```
SQL> select file_name, tablespace_name, bytes, status from dba_data_files;

FILE_NAME
--------------------------------------------------------------------------------
TABLESPACE_NAME                     BYTES STATUS
------------------------------ ---------- ---------
/u01/app/oracle/oradata/SADB/users01.dbf
USERS                             5242880 AVAILABLE

/u01/app/oracle/oradata/SADB/undotbs01.dbf
UNDOTBS1                        356515840 AVAILABLE

/u01/app/oracle/oradata/SADB/system01.dbf
SYSTEM                          943718400 AVAILABLE


FILE_NAME
--------------------------------------------------------------------------------
TABLESPACE_NAME                     BYTES STATUS
------------------------------ ---------- ---------
/u01/app/oracle/oradata/SADB/sysaux01.dbf
```

Step 7: parameter undo_retention

- Show the value of parameter undo_retention then change the value of it to be 3 minutes and gurantee this duration.

1. Show the current value:

```
SQL> show parameter undo_retention

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
undo_retention                       integer     900
SQL>
```

2. Change the value to 3 minutes (180 seconds):

```
SQL> alter system set undo_retention=180 ;

System altered.
```

3. Gurantee:

- Without Guarantee (Default): Oracle treats your 3 minutes as a suggestion. If it runs out of space, it overwrites your old data to keep the database moving.
- With Guarantee: Oracle treats your 3 minutes as a strict law. If it runs out of space, it fails new database changes (user gets an "out of space" error) rather than breaking the 3-minute rule.

- undo tablespace name that oracle created by default

```
SQL> SELECT FILE_NAME, TABLESPACE_NAME FROM DBA_DATA_FILES;

FILE_NAME
--------------------------------------------------------------------------------
TABLESPACE_NAME
------------------------------
/u01/app/oracle/oradata/SADB/users01.dbf
USERS

/u01/app/oracle/oradata/SADB/undotbs01.dbf
UNDOTBS1
```

- so to gurantee retention:

```
SQL> alter tablespace undotbs1 retention guarantee;

Tablespace altered.
```

Step 8: Create new undo tablespace

- reate new undo tablespace with name "NEWUNDO" with size 100 MB, and make it the default undo tablespace for the database instance.

```
SQL> create undo tablespace newundo datafile '/u01/app/oracle/oradata/SADB/newundo01.dbf' size 100m;

Tablespace created.

SQL> alter system set undo_tablespace = newundo ;

System altered.

SQL>
```

Step 9: View the redo log files

- view the redo log files names, size, group number, status, archived or not and their sizes in MB using the views "v$log" and "v$logfile".

```
SQL> select f.member , l.bytes/1024/1024 AS size_in_mb, l.group#, l.status, l.archived from v$log l, v$logfil
e f where l.group# = f.group#;

MEMBER
--------------------------------------------------------------------------------
SIZE_IN_MB     GROUP# STATUS           ARC
---------- ---------- ---------------- ---
/u01/app/oracle/oradata/SADB/redo03.log
       200          3 CURRENT          NO

/u01/app/oracle/oradata/SADB/redo02.log
       200          2 INACTIVE         NO

/u01/app/oracle/oradata/SADB/redo01.log
       200          1 INACTIVE         NO
```